INCH-POUND
MIL-PRF-55365/4L
28 August 2023
SUPERSEDING
MIL-PRF-55365/4K
13 March 2020

PERFORMANCE SPECIFICATION SHEET

CAPACITOR, CHIP, FIXED, TANTALUM, POLARIZED, ESTABLISHED RELIABILITY, NON-ESTABLISHED RELIABILITY, AND HIGH RELIABILITY, STYLES CWR06 AND CWR09

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-55365.

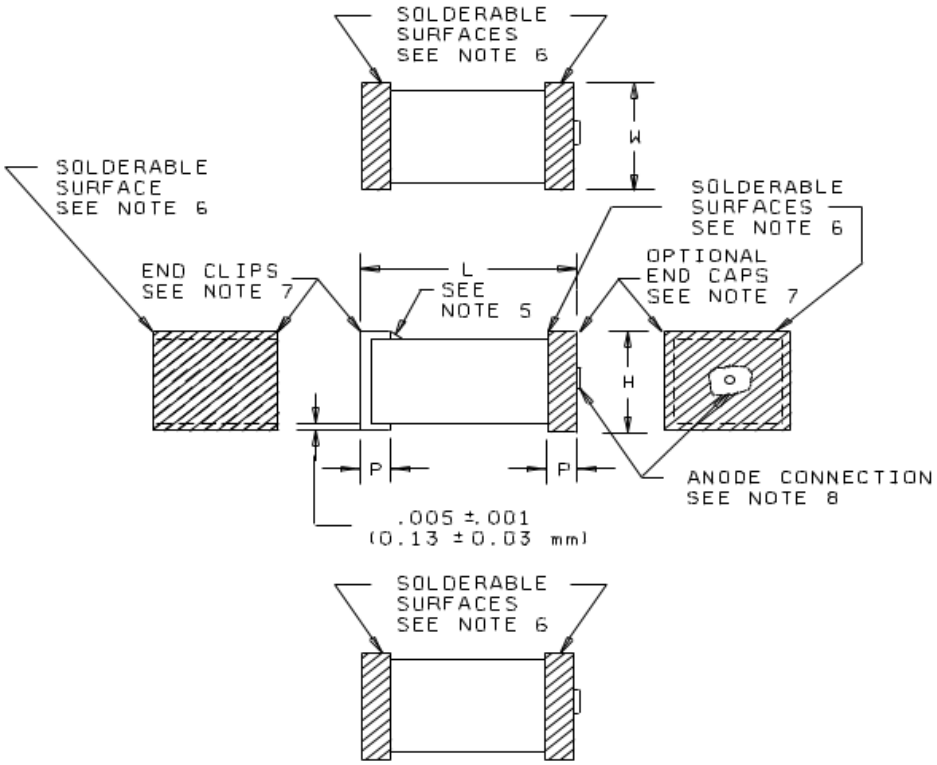


FIGURE 1. Style CWR06 (conformally coated) capacitors.

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Case size	L ± .015 (0.38)	W ± .015 (0.38)	H ± .015 (0.38)	P ± .005 (0.13)
A	.100 (2.54)	.050 (1.27)	.050 (1.27)	.030 (0.76)
B	.150 (3.81)	.050 (1.27)	.050 (1.27)	.030 (0.76)
C	.200 (5.08)	.050 (1.27)	.050 (1.27)	.030 (0.76)
D	.150 (3.81)	.100 (2.54)	.050 (1.27)	.030 (0.76)
E	.200 (5.08)	.100 (2.54)	.050 (1.27)	.030 (0.76)
F	.220 (5.59)	.135 (3.43)	.070 (1.78)	.030 (0.76)
G	.265 (6.73)	.110 (2.79)	.110 (2.79)	.050 (1.27)
H	.285 (7.24)	.150 (3.81)	.110 (2.79)	.050 (1.27)
X	.273 (6.93)	.213 (5.41)	.108 (2.74)	.047 (1.19)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in millimeters and are given for general information only.
3. Metric equivalents are in parentheses.
4. These capacitors are designed for mounting by dip soldering, thermo-compression bonding, reflow soldering, or other conventional means.
5. Cathode terminal may be epoxy coated on one surface for polarity identification.
6. Solderable surfaces are only those surfaces designated as such.
7. Anode and cathode terminals may be two-sided through five-sided, and may not be the same on some designs, but will meet all specification requirements.
8. The anode terminal shall be identified by the riser wire connection, which may extend the case size .015 inch (0.38 mm) maximum. The riser wire weld area is not solderable. The riser wire connection shall have a continuous weld of a minimum of 25 percent of the contact surface.
9. When solder coated terminations are required, add an additional .015 inch (0.38 mm) to the above listed tolerances for “L”, “W”, “H”, and “P” for each case size.

FIGURE 1. Style CWR06 (conformally coated) capacitors - Continued.

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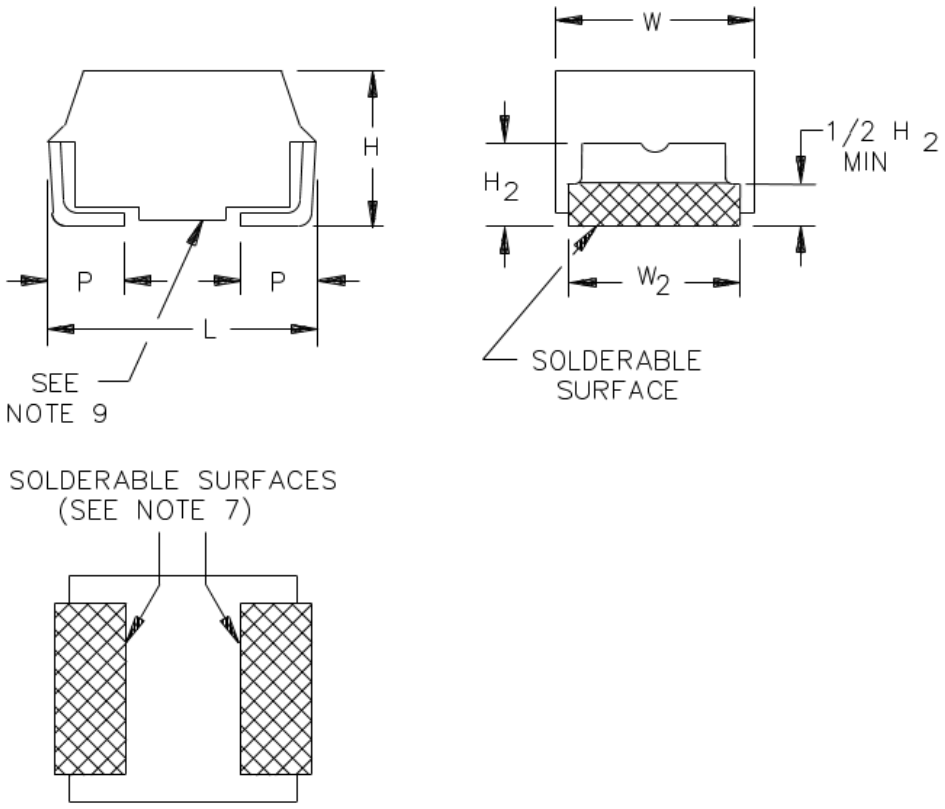


FIGURE 2. Style CWR09 (molded chip) capacitors.

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Case size	L ± .015 (0.38)	W ± .015 (0.38)	H ± .015 (0.38)	P + .010 (0.25) - .005 (0.13)	W_2	H_2 min
A	.100 (2.54)	.050 (1.27)	.050 (1.27)	.030 (0.76)	.050 ± .005 (1.27 ± .013)	.030 (0.76)
B	.150 (3.81)	.050 (1.27)	.050 (1.27)	.030 (0.76)	.050 ± .005 (1.27 ± .013)	.030 (0.76)
C	.200 (5.08)	.050 (1.27)	.050 (1.27)	.030 (0.76)	.050 ± .005 (1.27 ± .013)	.030 (0.76)
D	.150 (3.81)	.100 (2.54)	.050 (1.27)	.030 (0.76)	.095 + .005, -.010 (2.41 + 0.13, - 0.25)	.030 (0.76)
E	.200 (5.08)	.100 (2.54)	.050 (1.27)	.030 (0.76)	.095 +.005, - .010 (2.41 + 0.13, - 0.25)	.030 (0.76)
F	.220 (5.59)	.135 (3.43)	.070 (1.78)	.030 (0.76)	.130 ± .005 (3.30 ± 0.13)	.040 (1.02)
G	.265 (6.73)	.110 (2.79)	.110 (2.79)	.050 (1.27)	.105 ± .005 (2.67 ± 0.13)	.060 (1.52)
H	.285 (7.24)	.150 (3.81)	.110 (2.79)	.050 (1.27)	.145 + .005, - .020 (3.68 + 0.13, - 0.51)	.060 (1.52)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in millimeters and are given for general information only.
3. Metric equivalents are in parentheses.
4. These capacitors are designed for mounting by dip soldering, thermo-compression bonding, reflow soldering, or other conventional means.
5. The anode (+) terminal shall be identified by a marking on the case.
6. The termination width (W_2) shall be maintained to the top of the solderable area.
7. Solderable surfaces are only those surfaces designated as such. Termination edges are not considered solderable.
8. When solder coated terminations are required, add an additional .015 inch (0.38 mm) to the above listed tolerances for “L”, “H”, “P”, and “W_2” for each case size.
9. Shown with optional glue pad. At the option of the manufacturer, a glue pad between the solderable surfaces may be substituted.

FIGURE 2. Style CWR09 (molded chip) capacitors – Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and figure 2.

Termination finish: In accordance with termination finishes B, C, H, or K of MIL-PRF-55365.

DC rated voltage: See table I. Above +85°C, voltage derating is required (see MIL-PRF-55365).

Operating temperature range: -55°C to +125°C.

Product level designator: In accordance with MIL-PRF-55365.

DC leakage (DCL): See table I.

Capacitance: See table I.

Capacitance tolerance: ± 5 percent (J), ± 10 percent (K), or ± 20 percent (M).

Dissipation factor (DF): See table I.

Equivalent series resistance (ESR) at 100 kHz: In accordance with MIL-PRF-55365 and table I.

Resistance to soldering heat: In accordance with MIL-PRF-55365.

Stability at low and high temperatures: In accordance with MIL-PRF-55365.

Surge voltage: In accordance with MIL-PRF-55365.

Life:

2,000 hours: In accordance with MIL-PRF-55365

10,000 hours: In accordance with MIL-PRF-55365

Solderability: In accordance with MIL-PRF-55365, except that following steam aging; test samples may have a 30 minute bake out at +150°C prior to solder dipping.

Marking: Molded capacitors shall be marked on the top of the case in accordance with MIL-PRF-55365 as follows:

Case code A

J	(+)	polarity stripe, J = JAN
105X	-	capacitance in picofarad code and manufacturer's identifier

Case codes B, C, D, E, F, G, and H

J	(+)	polarity stripe, J = JAN
105	-	capacitance in picofarad code
35X	-	rated voltage and manufacturer's identifier

Alternate case code A marking

X	(+)	polarity stripe, X = manufacturer's identifier
105J	-	capacitance in picofarad code; J = JAN

Print orientation is optional. Additional marking may appear provided that it does not interfere with the required marking. At the option of the manufacturer, the lot date code may be marked on the top of the case for any case code. At the option of the manufacturer, the marking requirement for case codes B through H may be used for case code A.

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TABLE I. Styles CWR06 and CWR09 characteristics.

Part or identifying number (PIN) 1/ 2/	DC rated voltage (+85°C)	Cap. (nom)	DC Leakage (max)			Dissipation factor (max)			Max ESR 100 kHz +25°C Style CWR06	Max ESR 100 kHz +25°C Style CWR09	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C			
	Volts	µF	µA	µA	µA	%	%	%	Ohms	Ohms	
CWR0-C-225 - - -	4	2.2	1.0	10	12	6	8	8	8.0	8.0	A
CWR0-C-475 - - -	4	4.7	1.0	10	12	6	8	8	8.0	8.0	B
CWR0-C-685 - - -	4	6.8	1.0	10	12	6	8	8	5.5	5.5	C
CWR0-C-106 - - -	4	10.0	1.0	10	12	8	8	10	4.0	4.0	D
CWR0-C-156 - - -	4	15.0	1.0	10	12	8	10	12	3.5	3.5	E
CWR0-C-336 - - -	4	33.0	2.0	20	24	8	10	12	2.2	2.2	F
CWR0-C-686 - - -	4	68.0	3.0	30	36	10	12	12	1.1	1.1	G
CWR0-C-107 - - -	4	100.0	4.0	40	48	10	12	12	.9	.9	H
CWR0-D-155 - - -	6	1.5	1.0	10	12	6	8	8	8.0	8.0	A
CWR0-D-335 - - -	6	3.3	1.0	10	12	6	8	8	8.0	8.0	B
CWR0-D-475 - - -	6	4.7	1.0	10	12	6	8	8	5.5	5.5	C
CWR0-D-685 - - -	6	6.8	1.0	10	12	6	8	8	4.5	4.5	D
CWR0-D-106 - - -	6	10.0	1.0	10	12	8	10	12	3.5	3.5	E
CWR0-D-226 - - -	6	22.0	2.0	20	24	8	10	12	2.2	2.2	F
CWR0-D-476 - - -	6	47.0	3.0	30	36	10	12	12	1.1	1.1	G
CWR0-D-686 - - -	6	68.0	4.0	40	48	10	12	12	.9	.9	H
CWR0-F-105 - - -	10	1.0	1.0	10	12	6	8	8	12.0	10.0	A
CWR0-F-225 - - -	10	2.2	1.0	10	12	6	8	8	8.0	8.0	B
CWR0-F-335 - - -	10	3.3	1.0	10	12	6	8	8	5.5	5.5	C
CWR0-F-475 - - -	10	4.7	1.0	10	12	6	8	8	4.5	4.5	D
CWR0-F-685 - - -	10	6.8	1.0	10	12	6	8	8	3.5	3.5	E
CWR0-F-156 - - -	10	15.0	2.0	20	24	8	8	10	2.5	2.5	F
CWR0-F-336 - - -	10	33.0	3.0	30	36	10	12	12	1.1	1.1	G
CWR0-F-476 - - -	10	47.0	5.0	50	60	10	12	12	.9	.9	H
CWR0-H-684 - - -	15	.68	1.0	10	12	6	8	8	12.0	12.0	A
CWR0-H-155 - - -	15	1.5	1.0	10	12	6	8	8	8.0	8.0	B
CWR0-H-225 - - -	15	2.2	1.0	10	12	6	8	8	5.5	5.5	C
CWR0-H-335 - - -	15	3.3	1.0	10	12	6	8	8	5.0	5.0	D
CWR0-H-475 - - -	15	4.7	1.0	10	12	6	8	8	4.0	4.0	E
CWR0-H-106 - - -	15	10.0	2.0	20	24	6	8	8	2.5	2.5	F
CWR0-H-226 - - -	15	22.0	4.0	40	48	6	8	8	1.1	1.1	G
CWR0-H-336 - - -	15	33.0	5.0	50	60	8	8	10	.9	.9	H
CWR0-J-474 - - -	20	.47	1.0	10	12	8	8	10	16.0	14.0	A
CWR0-J-684 - - -	20	.68	1.0	10	12	6	8	8	14.0	10.0	B
CWR0-J-105 - - -	20	1.0	1.0	10	12	6	8	8	12.0	12.0	B
CWR0-J-155 - - -	20	1.5	1.0	10	12	6	8	8	6.0	6.0	C
CWR0-J-225 - - -	20	2.2	1.0	10	12	6	8	8	5.0	5.0	D
CWR0-J-335 - - -	20	3.3	1.0	10	12	6	8	8	4.0	4.0	E
CWR0-J-685 - - -	20	6.8	2.0	20	24	6	8	8	2.4	2.4	F
CWR0-J-156 - - -	20	15.0	3.0	30	36	6	8	8	1.1	1.1	G
CWR0-J-226 - - -	20	22.0	4.0	40	48	6	8	8	.9	.9	H
CWR0-K-334 - - -	25	.33	1.0	10	12	6	8	8	15.0	15.0	A
CWR0-K-684 - - -	25	.68	1.0	10	12	6	8	8	10.0	7.5	B
CWR0-K-105 - - -	25	1.0	1.0	10	12	6	8	8	6.5	6.5	C
CWR0-K-155 - - -	25	1.5	1.0	10	12	6	8	8	6.5	6.5	D
CWR0-K-225 - - -	25	2.2	1.0	10	12	6	8	8	3.5	3.5	E
CWR0-K-475 - - -	25	4.7	2.0	20	24	6	8	8	2.5	2.5	F
CWR0-K-685 - - -	25	6.8	2.0	20	24	6	8	8	1.2	1.2	G
CWR0-K-106 - - -	25	10.0	3.0	30	36	6	8	8	1.4	1.4	G
CWR0-K-156 - - -	25	15.0	4.0	40	48	6	8	8	1.0	1.0	H

See footnotes at the end of table.

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TABLE I. Styles CWR06 and CWR09 characteristics - Continued.

PIN 1/ 2/	DC rated voltage (+85°C)	Cap. (nom)	DC Leakage (max)			Dissipation factor (max)			Max ESR 100 kHz +25°C Style CWR06	Max ESR 100 kHz +25°C Style CWR09	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C			
	Volts	µF	µA	µA	µA	%	%	%	Ohms	Ohms	
CWR0-M-224 - - -	35	.22	1.0	10	12	6	8	8	24.0	18.0	A
CWR0-M-474 - - -	35	.47	1.0	10	12	6	8	8	17.0	10.0	B
CWR0-M-684 - - -	35	.68	1.0	10	12	6	8	8	10.0	8.0	C
CWR0-M-105 - - -	35	1.0	1.0	10	12	6	8	8	6.5	6.5	D
CWR0-M-155 - - -	35	1.5	1.0	10	12	6	8	8	4.5	4.5	E
CWR0-M-335 - - -	35	3.3	1.0	10	12	6	8	8	2.5	2.5	F
CWR0-M-475 - - -	35	4.7	2.0	20	24	6	8	8	1.5	1.5	G
CWR0-M-685 - - -	35	6.8	3.0	30	36	6	8	8	1.3	1.3	H
CWR0-N-104 - - -	50	.10	1.0	10	12	6	8	8	22.0	22.0	A
CWR0-N-154 - - -	50	.15	1.0	10	12	6	8	8	25.0	17.0	A
CWR0-N-224 - - -	50	.22	1.0	10	12	6	8	8	17.0	14.0	B
CWR0-N-334 - - -	50	.33	1.0	10	12	6	8	8	12.0	12.0	B
CWR0-N-474 - - -	50	.47	1.0	10	12	6	8	8	8.0	8.0	C
CWR0-N-684 - - -	50	.68	1.0	10	12	6	8	8	7.0	7.0	D
CWR0-N-105 - - -	50	1.0	1.0	10	12	6	8	8	6.0	6.0	E
CWR0-N-155 - - -	50	1.5	1.0	10	12	6	8	8	4.0	4.0	F
CWR0-N-225 - - -	50	2.2	2.0	20	24	6	8	8	2.5	2.5	F
CWR0-N-335 - - -	50	3.3	2.0	20	24	6	8	8	2.0	2.0	G
CWR0-N-475 - - -	50	4.7	3.0	30	36	6	8	8	1.5	1.5	H

1/ Complete PIN shall include additional symbols to indicate style, termination finish, capacitance tolerance, product level designator, and, if applicable, surge current option letter. If optional surge current is not required, the last "-" shall be deleted.

2/ Styles CWR06 and CWR09 are, with the acquiring agency approval, considered to be interchangeable, provided the size, capacitance, capacitance tolerance, failure rate level, rated voltage, maximum ESR value, and surge current option (if applicable) remain the same.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2023-027)

Review activities:
Army - AR, MI
Navy - AS, MC, OS, SH
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